	m_1^z $[M_\odot]$	m_2^z $[M_\odot]$	D_L [Mpc]	ι [rad]	$f_{\rm ref}$ [Hz]	ra [rad]	dec [rad]	ψ [rad]	Detector	$t_{\rm gps}$ [s]	SNR
Base Signal	20	10	500	0.2	20	4.67	0.65	2.34	Hanford	1192529720	16.78

TABLE I. Injection Parameters for the base signal. In the high-SNR case, the luminosity distance is changed, now taking a value of 100 Mpc instead and accordingly, the SNR changes to 83.91. Note that the SNR values quoted here refer to the non-modified injection, for the modified one the SNR will be slightly different (with deviations being on the order of 1%). We use priors that are uniform in component mass for $\mathcal{M}^z, q$, uniform in $\cos(\iota)$ for the inclination (i.e. $p(\iota) \propto \sin(\iota)$), uniform in comoving volume for the luminosity distance, and uniform in coalescence time t_c (no prior for ϕ_0 because we use a marginalized-phase likelihood).

Configuration		$\Delta\mathcal{M}^z$ $[M_\odot]$		$\sigma_{\mathcal{M}^z}$ $[M_\odot]$		Δq		σ_q	
		PE	Fisher	PE	Fisher	PE	Fisher	PE	Fisher
$\delta\tilde{A} = \delta\tilde{\phi} = 0.004$	Base Signal	−0.023	−0.026	0.040	0.039	−0.012	0.001	0.143	0.137
	high-SNR	−0.028	−0.026	0.0077	0.0078	0.007	0.001	0.0271	0.0273
$\delta\tilde{A} = \delta\tilde{\phi} = 0.01$	Base Signal	−0.064	−0.04	0.040	0.039	−0.0048	−0.057	0.143	0.137
	high-SNR	−0.069	−0.039	0.0078	0.0078	0.013	−0.058	0.0275	0.0273

TABLE II. Summary of results from full PE runs and Fisher matrix estimates. The numbers one should primarily compare the neighboring columns labeled "PE" and "Fisher", for each of the four quantities that appear in the upper row. Moreover, when comparing them one should keep in mind that neither number is expected to be perfectly accurate, explaining certain small deviations; PE results are always subject to sampling uncertainties (which are noticeable for instance for Δq), while limitations of the Fisher approach have been laid out at the beginning of this section and Appendix A.

corresponds to the to data from next-generation GW detectors, where a tenfold increase in sensitivity is aimed for. Here, Fisher estimate and true recovery are basically fully separate. All of this plays into why we choose to not use Fisher matrix estimates for the rest of this paper. A suitable replacement to still account for these systematic errors by using waveform with an error model like (13) on top of them is discussed in the next section.

V. SIMULATIONS AND TESTS OF PARAMETER ESTIMATION FRAMEWORK

This section uses simulations to test the WF-Error parametrizations described in section II A. Though we do not have access to the 'true' waveform model, we can still check the validity of the PE framework we describe by considering a reference waveform model, $h_{\rm ref}$, modifying the $h_{\rm ref}$, and checking if our parametrization can pick up the introduced modification. We take a reference signal $h_{\rm ref}$ described by a waveform approximant and introduce errors in the model described by parametrizations in equations (12) and (13). Now we have a modified signal $h_{\rm mod}$, which shall have a systematic error compared to the $h_{\rm ref}$. We consider $h_{\rm mod}$ as the 'true' signal and inject it in the strain data of LIGO detectors at Hanford (H1), Livingston (L1), and Virgo detector (V1). Then, we use the Bayesian PE data analysis pipeline to recover the source properties using $h_{\rm ref}$ as our waveform model with and without incorporating waveform uncertainties in PE. We use non-spinning signals (spin parameters in waveform approximant set to zero) with zero noise, except the last subsection where we deal with missing physics. In the subsequent subsections, we describe each injection-recovery campaign's injected signals and PE scheme.

For PE, we use the publicly available toolkit PyCBC Inference [89], and for sampling the likelihood function, we use the nested sampling algorithm implemented in the publicly available code Dynesty [113]. We also implement WF-Error parametrizations in code that serves as a plugin [114] for PyCBC. We use the aLIGOZeroDetHighPower PSD for the LIGO Hanford and Livingston detectors and the AdvVirgo PSD for the Virgo detector. These PSD functions are implemented in publicly available modules LALSimulation, which is a part of the LALSuite code [115]. We use two sets of runs for each injection:

- In first set of run, we sample the parameter space with the following parameters: source frame chirp mass ($\mathcal{M}^{\rm src}$), mass ratio $q(q = \frac{m_1}{m_2}, m_1 \geq m_2)$, comoving volume, inclination angle, and geocentric time of arrival of the signal (t_c). We fix right ascension (RA), declination (dec), and polarization angle (ψ) for these runs.
- In the second sets of runs, in addition to the parameters described above, we also vary ra, dec, and ψ.

We transform the sampling parameters to the waveform model parameters, such as detector frame masses

Recovery Model	Sampling Parameters	Description	Prior
IMRPhenomPv2	$\mathcal{M}^{\rm src}$	Source frame chirp mass	Uniform in component masses
	$q = \frac{m_2}{m_1}$ $(m_1 \geq m_2)$	Mass ratio	
	V_c	Comoving volume	Uniform: $\pi(V_c) = \mathcal{U}(5 \times 10^3, 9 \times 10^9)$
	ι	Inclination angle	$\pi(\iota) \propto \sin(\iota)$
	t_c	Time of arrival	Uniform: $\pi(t_c) = \mathcal{U}(t_c - 0.1, t_c + 0.1)$
	RA, dec	Right ascension and declination	Uniform Sky
	ψ	Polarization Angle	Uniform angle: $\pi(\psi) = \mathcal{U}(0, 2\pi)$
IMRPhenomPv2 + WF Error	$\delta A_i, \delta\phi_i$	WF-Error parameters	$\mathcal{N}(\mu = 0, \sigma = 0.032)$

TABLE III. This table describes the priors used in the PE of simulated signals. The distribution $\mathcal{U}(a, b)$ refers to a one-dimensional uniform distribution defined within the interval (a, b). For the WF-Error parameters, when we employ cubic spline curves with ten nodal points across the frequency range of $(20, 500)$ Hz. The prior distributions for each $\delta\tilde{A}_i$ and $\delta\tilde{\phi}_i$ are specified accordingly. The same prior is used for the special test cases, when we do not use cubic spline but a constant shift in amplitude and phase with parameters $\delta\tilde{A}, \delta\tilde{\phi}$. $\pi(\theta)$ denotes the prior distribution used for a parameter θ.

(m_1^z, m_2^z), and luminosity distance (D_L) through standard transformations. We fix the spins to be zero for the study in this manuscript, except for in sub-section V C. For the PE runs described by WF-Error parametrizations, we use additional parameters. We use two additional WF-Error parameters for the constant shift models: $\delta\tilde{A}$ and $\delta\tilde{\phi}$. For the cubic-spline model, we use ten frequency nodal points with log spacing between a minimum frequency of 20 Hz and a maximum frequency of 500 Hz.

In Table III, we describe the prior distribution for each of the abovementioned parameters. All the injections described in this section are zero-noise injections.

A. Simplest modification: A constant shift in amplitude and phase with rel-phase parametrization

We consider a GW1500915-like signal with source frame masses $m_1^{\rm src}, m_2^{\rm src} = (36\ M_\odot,\ 29\ M_\odot)$ at a distance of 500 Mpc. We generate the corresponding reference signal using the `IMRPhenomPv2` waveform model and introduce the errors in reference signal with rel-phase parametrization corresponding to:

$$(\delta\tilde{A}, \delta\tilde{\phi}_{\rm rel}) = (0.01, 0.01) \tag{19}$$

For this most straightforward case, a constant phase shift is expected to be significant for rel-phase parametrization. For abs-phase parametrization, a constant phase shift will not introduce any physical effect except a constant shift in time. We will consider the general modification with abs-phase parametrization in the following subsections. We perform PE with the injected signal in the detector network, HLV, with the following combination:

- **Injection: standard, PE: standard** In this scenario, we inject the reference signal as described above and perform the PE with the same waveform model without introducing the WF-Error parametrization.
- **Injection: modified, PE: standard** In this scenario, we inject the modified signal and perform the PE with the reference waveform model without introducing the waveform errors.
- **Injection: modified, PE: WF-Errors** In this scenario, we inject the modified signal, perform the PE with the reference waveform model + WF-Error parametrization.
- **Injection: standard, PE: WF-Errors** Here, we inject the reference signal, perform the PE with the reference waveform model+WF-Error.

In Figure 5, we show the injection and recovery of various combinations. We show that when we have a modified injection (equivalent to a systematic error with rel-phase parametrization), and we perform PE with reference waveform model, we get a bias in the detector frame chirp mass $\mathcal{M}^{\rm z}$. This bias is corrected when we use the right parametrization to account for waveform systematics. We also note that we can correct the bias when we use cubic splines for constant shift error in the reference signal. However, it is to be noted that when we use abs-phase parametrization to correct for the bias introduced by rel-phase parametrization, it fails to correct for the bias. These effects are more visible in a louder signal, which is simulated by injection at a closer luminosity distance (100) Mpc compared to one at 500 Mpc. We also show the network SNR, $\rho^{\rm net} = \sqrt{\sum_i \rho_i^2}$, for the reference waveform injection and recovery, where ρ_i is the SNR for the i-th detector. Figure 6 shows the recovery of parameter $\delta\tilde{\phi}$ with rel-phase parametrization. We show that we can recover the injected value of $\delta\tilde{\phi}$ with the correct parametrization. When there is no modification to the

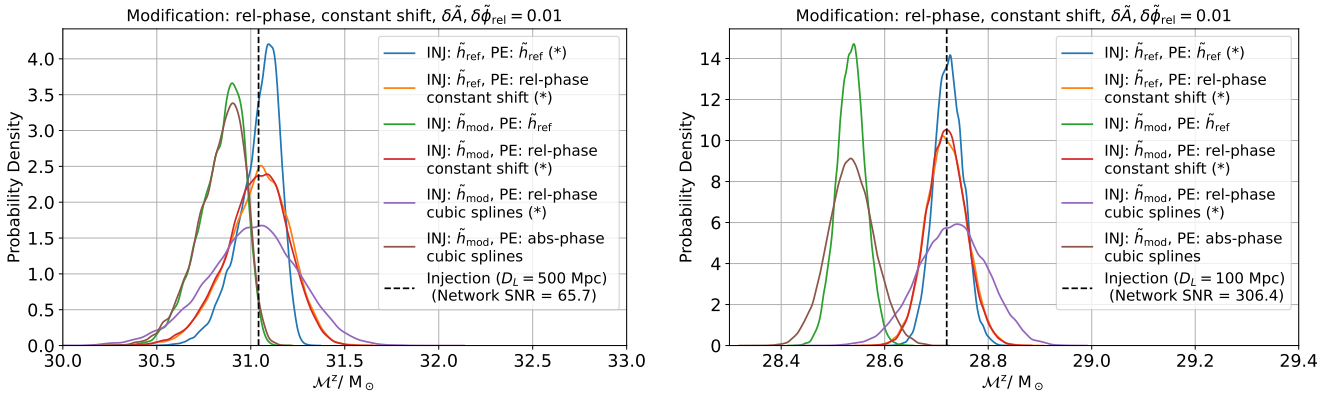

FIG. 5. Injection and recovery are shown for a non-spinning GW150914-like signal with source frame masses $(m_1^{src}, m_2^{src}) = (36, 29)\ M_\odot$, at a luminosity distance of 500 Mpc (left panel) and a distance of 100 Mpc (right panel). We also inject a modified signal with rel-phase modification with parameters $(\delta\tilde{A}, \delta\tilde{\phi}) = (0.01, 0.01)$. When we inject a modified signal and use the incorrect model to recover, we see a bias in the recovery of detector frame chirp mass $\mathcal{M}^z$. We also show that using the correct parametrization (in this case, rel-phase) can correct the bias and get a broader marginalized posterior sample. The bias is more prominent in a louder signal (right panel) than in a comparatively weaker signal (left panel). We mark that combination of injection and recovery by an asterisk ($\star$) where we do not expect a bias. The vertical dashed line represents the injected value of $\mathcal{M}^z$. For these PE runs, we fix the parameters RA, dec, and polarization to the injected values. The network SNR for reference injection is shown in each panel.

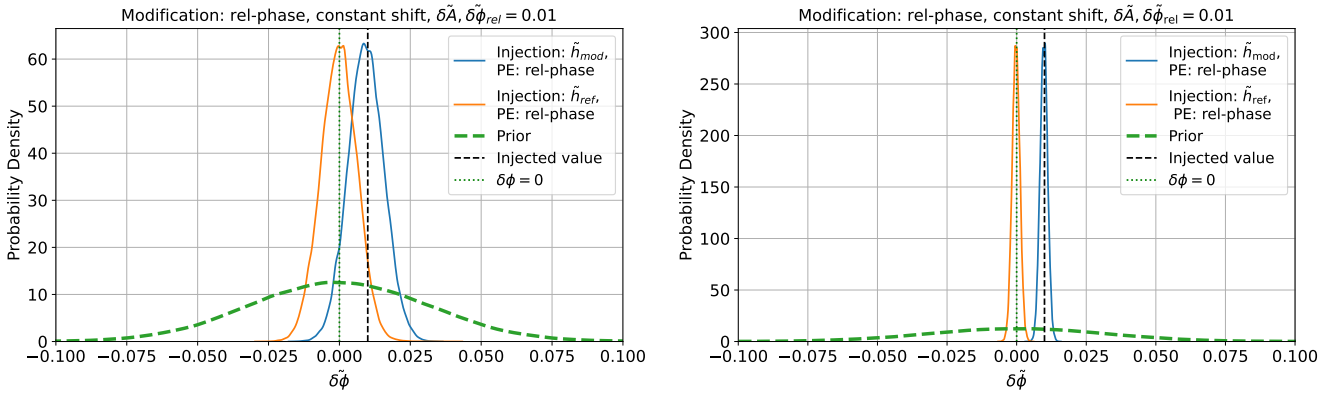

FIG. 6. Recovery of the parameter $\delta\tilde{\phi}$ for rel-phase parametrization for various scenarios of injected parameters. The signal injected here is a non-spinning GW150914-like, as in Figure 5. The left panel has an injection at 500 Mpc, while the right panel shows the case with an injection at 100 Mpc. The vertical dashed line shows the injected value for $\delta\tilde{\phi}$ for the modified signal. The green dashed curve shows the prior distribution for the $\delta\tilde{\phi}$ parameter. We can recover the injected value of $\delta\tilde{\phi}$ with the correct parametrization for PE. We recover $\delta\tilde{\phi} = 0$ for the reference injection. For these PE runs, we fix the parameters RA, dec, and polarization to the injected values.

reference signal, the rel-phase parametrization recovers $\delta\phi = 0$. We fix the RA, dec, and ψ for the runs shown in figures 5 and 6. We find that the $\delta\tilde{A}$ parameter returns the prior distribution and does not have a constraining power for the PE runs described in this subsection. For simplistic cases, we expect $\delta\tilde{A}$ to be degenerate with distance, i.e., a small constant shift in the amplitude can be modeled in the luminosity distance. At this stage, it can be argued that we can get rid of $\delta\tilde{A}$ parameter in the WF-Error parametrization and focus on $\delta\tilde{\phi}$ parameter. However, we shall keep this parameter in our model for more general scenarios where errors in amplitude can be a function of the frequency. We do not find any bias in the other parameters considered in the simulations in this subsection, except for the detector frame chirp mass.

Through these simulations, we have established that for a simplistic model, a systematic error corresponding to $\delta\tilde{\phi} = 0.01$ can lead to significant bias in the detector frame chirp mass for GW150914-type signal, with the detector sensitivity of advanced LIGO design sensitivity. We have also shown that, by using WF-Error